3/8/2026

4. Setting Up and Configuration of LAN

Aim :-

To design, setup, and configure a LAN connecting four computers through a network switch, assign static IP addr to all device verify connectivity using the ping command and demonstrate resorce / data accross the network.

Equipments Required :-

i). Personal Computer (4)

ii). Network switch (1)

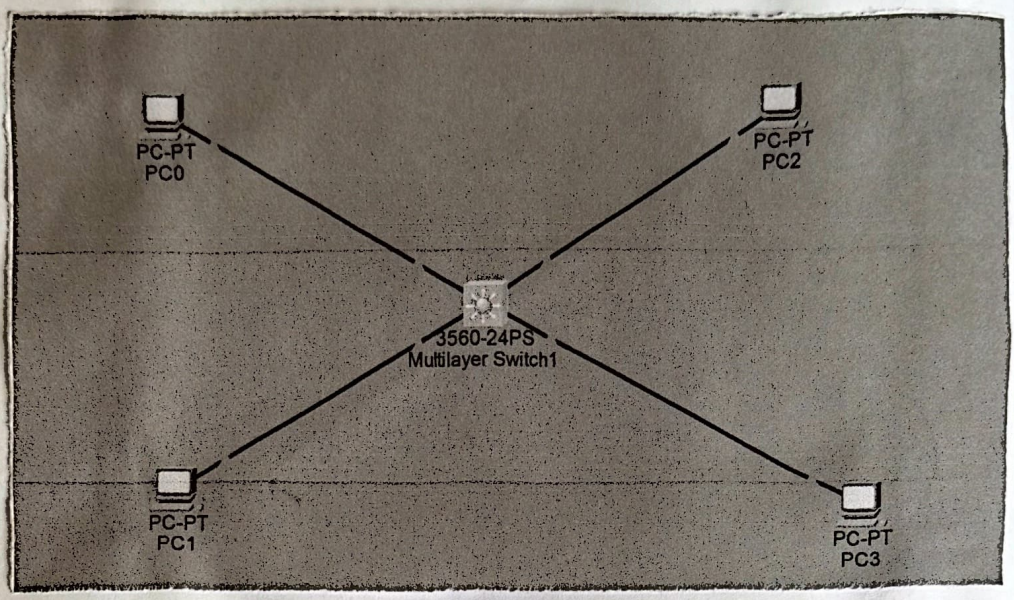
iii). Ethernet Cable (Straight through)

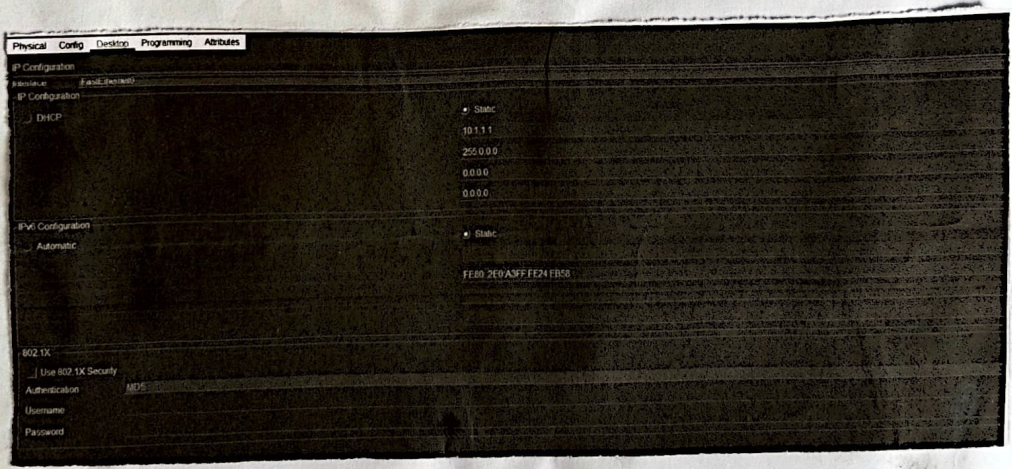
Network diagram :-

The LAN uses star topology.

IP Address Configuration table :-

Device	IP addr	Subnet Mask	Connected to
PC0	10.1.1.1	255.0.0.0	Port Fa0/1
PC1	10.1.1.2	255.0.0.0	Port Fa0/2
PC2	10.1.1.3	255.0.0.0	Port Fa0/3
PC3	10.1.1.4	255.0.0.0	Port Fa0/4
Switch	10.1.1.5	255.00.0	-


PC-PT
PC0
PC-PT
PC2
3560-24PS
Multilayer Switch1
PC-PT
PC1
PC-PT
PC3


Physical
Config
Desktop
Programming
Attributes
IP Configuration
IP Configuration
DHCP
Static
10.1.1.1
255.0.0.0
0.0.0.0
0.0.0.0
IPv6 Configuration
Automatic
Static
FE80::2E0:A3FF:FE24:FB58
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

Implementation in Cisco Packet Tracer:-

1. Open packet tracer app and search for the required device
2. Select Network section and under network section choose the switch (Eg:- 3560-24PS)
3. Select device section and select 4 Pc and place them in star topology.
4. Select cabel section and choose Copper Straigth cable and connect it to the fast ethernet port on both sides (end device to switch port)
5. Assign IP address to PC's
6. Configure switch in Proporatory OS of the CISCO network devices:-

```
enable
configure terminal
interface vlan1
ip address 10.1.1.5 255.000.000.000
no shutdown
exit
```

7. Verify Connection via ping Command testing.
8. Demonstrate data sharing using ARP or any other such kind of protocol.

Result :-

All four PC's were successfully connected to switch in a star ~~protocol~~ topology and tested using ping protocol and hence the connection and communication is tested for correctness